

A3 Report	Title of Issue:	Stryten Plastic Collection and Recycling			CAR/PAR No:		Date Opened:	6/2/2025	
	A3 Leader:	Michael Hawk	Team:	Chris Williams			Date Closed:	7/24/2025	

Step 1: Define the Problem: (What, When, Where, First time observed, Total population)

The plant throws away all plastics with the rest of the trash instead of recycling, which increases the amount of trash that is sent to the landfill. It would be beneficial to weigh the plastics and send them to a recycling plant instead of sending them to the landfill.

Step 2: Problem Breakdown (Pictures, Drawings, Charts, etc. to illustrate problem as much as possible)

-The plastics are thrown away along with all of the trash, leading to a very large amount of trash going to the landfill each year
-If the plastics were weighed and sent to a recycling facility, that would greatly reduce the amount of trash the plant would contribute to the landfill

Current:

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graph LR
    A((Plastics Placed into Baler)) --> B((Compress into Bale))
    B --> C{Baler Full?}
    C -- Yes --> D((Send Bale to Landfill))
    C -- No --> A
  
```

ESG Friendly:

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graph LR
    A((Plastics Placed into Baler)) --> B((Compress into Bale))
    B --> C{Baler Full?}
    C -- Yes --> D((Weigh Bale))
    D --> E((Operator Records Data on Sheet))
    E --> F((Bale Sent to Recycling Plant))
    C -- No --> A
  
```

Step 3: Set the Goal/Target: (Define the desired situation once the problem is solved)

To reduce the weight of trash sent to the landfill by 54,000 lbs to fulfill ESG standards and improve plant environmental statistics.

Step 4: Root Cause Analysis (Identify possible root causes and validate their contribution to the problem)

Possible Cause	Why? Because...	Proposed Countermeasure to be implemented
Lack of process or tracking for the disposal of plastic from the plant	Why? It is easier to just send plastic to the landfill	Create and implement process for operators to weigh, report, and send the plastics to the recycling plant, as well as sustain this process after implementation is complete.
	Why? It costs less and requires less labor	
	Why? Sending plastic to the landfill requires more steps	
	Why? It separates the trash from the plastic and that takes labor	
	Why? It is a process that would decrease the total trash sent to the landfill	Responsible: Michael Hawk, Chris Williams

Step 5: Develop and Prioritize Countermeasures

Prioritize implementation of countermeasures by using the calculated value in Overall column = Cost score X Feasibility score X Effectiveness score. Prioritize Higher Overall First									
Score 3 = Low Cost, Easy to Implement, Effective Score 2 = Mid \$\$, long time to Implement, Some what Effective Score 1 = High \$\$\$, Hard to Implement, Low Effectiveness									
Cause No.	Root Cause	Countermeasure	Cost \$	X	Feasibility X	Effectiveness	=	Overall	Comments
1	Lack of Weighing for Plastics	Create process for operators to weigh the plastic	3		3	3		27	
2	Plastics are sent to the landfill	Send plastic to a recycling plant instead of the landfill	1		3	3		9	
3	No system in place for tracking plastic	Ensure that operators continually report weights and when they were done	3		3	3		27	

Step 6: Implement Countermeasures

ID	Countermeasure Implementation Task	Start	Finish	Responsible	Weeks after start date (Check the boxes for duration to complete task)											
					1	2	3	4	5	6	7	8	9	10	11	12
1	Implementing tracking sheet for operators	6/2/2025	6/2/2025	Michael H												
2	Have operators weigh plastic for data collection	6/9/2025	7/7/2025	CI												
3	Create process to send plastic to recycling	6/9/2025	7/24/2025	CI												
4																
5																
6																

Step 7: Countermeasure Effectiveness (Validate using data)

Set the Goal+limits	Weeks after start date (Trend the performance every week)											
	1	2	3	4	5	6	7	8	9	10	11	12
	Actual:											
- Validate Countermeasure(s) had the expected impact on the problem defined in Step 1. - Trend the actual levels weekly and compare to the Goal/Target defined in Step 3.												

Step 8a: Standardize Successes (If a question answer is "No", the appropriate task needs to be completed to answer "Yes")

No	Process Changes	Yes	No	N/A	No	Tools / Equipment Change	Yes	No
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